

```
import pandas as pd
import matplotlib.pyplot as plt

data = pd.DataFrame({
    "Property": ["P1", "P2", "P3", "P4", "P5", "P6", "P7", "P8", "P9", "P10", "P11", "P12"],
    "Price_Lakh": [85, 120, 95, 150, 70, 135, 110, 90, 160, 75, 125, 100],
    "Area_sqft": [1100, 1450, 1200, 1800, 950, 1600, 1300, 1050, 1900, 900, 1500, 1250],
    "Age_Years": [8, 15, 5, 20, 3, 18, 10, 6, 22, 4, 12, 7],
    "Days_On_Market": [210, 245, 190, 280, 75, 230, 200, 65, 300, 90, 215, 185],
    "Distance_City_km": [12, 18, 8, 25, 5, 20, 15, 7, 28, 6, 17, 10],
    "Condition_Score": [7, 6, 8, 5, 9, 6, 7, 9, 5, 8, 6, 8]
})

data["Unsold_6_Months"] = data["Days_On_Market"] > 180
data
```

	Property	Price_Lakh	Area_sqft	Age_Years	Days_On_Market	Distance_City_km	Condition_Score	Unsold_6_Months
0	P1	85	1100	8	210	12	7	True
1	P2	120	1450	15	245	18	6	True
2	P3	95	1200	5	190	8	8	True
3	P4	150	1800	20	280	25	5	True
4	P5	70	950	3	75	5	9	False
5	P6	135	1600	18	230	20	6	True
6	P7	110	1300	10	200	15	7	True
7	P8	90	1050	6	65	7	9	False
8	P9	160	1900	22	300	28	5	True
9	P10	75	900	4	90	6	8	False
10	P11	125	1500	12	215	17	6	True
11	P12	100	1250	7	185	10	8	True

Identify properties unsold for more than six months

```
unsold = data[data["Unsold_6_Months"]].copy()

print("Properties unsold for more than six months:")
print(unsold[["Property", "Price_Lakh", "Area_sqft", "Age_Years",
              "Days_On_Market", "Distance_City_km", "Condition_Score"]])

print("\nNumber of properties:", len(unsold))
```

```
Properties unsold for more than six months:
Property Price_Lakh Area_sqft Age_Years Days_On_Market \
0      P1         85     1100         8         210
1      P2        120     1450        15         245
2      P3         95     1200         5         190
3      P4        150     1800        20         280
5      P6        135     1600        18         230
6      P7        110     1300        10         200
8      P9        160     1900        22         300
10     P11        125     1500        12         215
11     P12        100     1250         7         185

Distance_City_km Condition_Score
0              12              7
1              18              6
2               8              8
3              25              5
5              20              6
6              15              7
8              28              5
10             17              6
11             10              8

Number of properties: 9
```

▼ Compare long-listed properties with properties sold within six months

```
summary = data.groupby("Unsold_6_Months")[
    ["Price_Lakh", "Area_sqft", "Age_Years", "Distance_City_km", "Condition_Score"]
].mean()

summary.index = ["Sold/within 6 months", "Unsold > 6 months"]
summary
```

	Price_Lakh	Area_sqft	Age_Years	Distance_City_km	Condition_Score
Sold/within 6 months	78.333333	966.666667	4.333333	6.0	8.666667
Unsold > 6 months	120.000000	1455.555556	13.000000	17.0	6.444444

```
plt.figure(figsize=(8, 5))
plt.bar(
    ["Within 6 months", "Unsold > 6 months"],
    [
        data.loc[~data["Unsold_6_Months"], "Price_Lakh"].mean(),
        data.loc[data["Unsold_6_Months"], "Price_Lakh"].mean()
    ]
)
plt.ylabel("Average Price (Lakh)")
plt.title("Average Property Price by Selling Duration")
plt.show()
```



```
plt.figure(figsize=(8, 5))
plt.scatter(data["Price_Lakh"], data["Days_On_Market"])
plt.xlabel("Price (Lakh)")
plt.ylabel("Days on Market")
plt.title("Price vs Days on Market")
plt.show()
```

